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Dear Editor,

I wish to submit a manuscript titled *Plants in limbo – the theory of detrital photosynthesis* for consideration as a *Theories* article in *Quantitative Plant Biology*. I confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

In this paper, I consolidate the theory of detrital photosynthesis, which has thus far only been treated vaguely in the literature. Specifically, I present some original data on photosynthesis in seagrass detritus, develop the first conceptual model of detrital photosynthesis and quantitatively compare its longevity and influence on decomposition across 92 species from all major plant groups.

My findings indicate that while all plants display some photosynthesis following detachment, macroalgae are unique among plants in exhibiting detrital photosynthesis over periods longer than a month. These timescales are sufficient to influence decomposition. Detrital photosynthesis therefore provides a novel mechanism by which light can slow decomposition, contrary to photodegradation. This contradicts conventional decomposition theory and highlights that many assumptions about macroalgal decomposition are wrong, which is especially relevant to the current blue carbon debate.

I am a quantitative plant ecologist, so greatly welcomed the inception of your journal with its broad botanical scope. I have been working on the topic of detrital photosynthesis for a few years now. After discovering your *Theories* article type, which I thought perfectly suited this unconventional topic, I tailored my analysis and manuscript to this article type. As such I “focus on new concepts” (detrital photosynthesis), “build on meta-analysis ... to extract overarching trends or question existing dogma thanks to quantitative analyses” (my meta-analysis questions conventional decomposition theory and blue carbon dogma) and “employ computational, theoretical, or analytical approaches to derive novel conceptual models that make quantitative predictions that are experimentally testable” (I present the first model of detrital photosynthesis and make concrete predictions that can be empirically tested).

All data and code are available in my open-access repository at github.com/lukaseamus/plant-limbo and I encourage you to pass these on to the reviewers to facilitate their review. Data collated for the meta-analysis are additionally deposited at github.com/lukaseamus/detrital-photosynthesis which is linked to a Zenodo data publication (doi.org/10.5281/zenodo.15054888).

Sincerely,
Luka Seamus Wright